ETALE LOCAL FREENESS ALONG THE FIBRE FOR A FINITE TYPE SHEAF FLAT OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a two step etale descent construction: affine etale localization of the base followed by an affine surjective etale cover, flatness transferred by base change, the scheme case lemma, and composition of the steps

Editorial note. The label lemma-finite-type-flat-along-fibre-free-variant names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Base change of flatness of a module along morphisms of algebraic spaces
- Existence of affine etale neighbourhoods of a point and of affine schemes surjective etale over a quasi-compact algebraic space
- The scheme case: the freeness variant of the finite type flat along the fibre lemma for schemes

Source reference. Original lemma 05L1: sources/stacks-chow/tags.txt, source lines 7220–7220. Prerequisite pointer invoked at proof line(s) 819; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05VW: sources/stacks-chow/tags.txt, source lines 7565–7565. Prerequisite pointer invoked at proof line(s) 816; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Let $y \in |Y|$. Set $F = f^{-1}(\{y\}) \subset |X|$. Assume that*

(1) *f is of finite type,*
(2) *$\mathcal{F}$ is of finite type, and*
(3) *$\mathcal{F}$ is flat over Y at all $x \in F$.*

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Then there exists an étale morphism $(Y', y') \to (Y, y)$ where Y' is a scheme and a commutative diagram of algebraic spaces

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ \downarrow & & \downarrow \\ Y & \longleftarrow & \mathrm{Spec}(\mathcal{O}_{Y',y'}) \end{array}$$

such that $X' \to X \times_Y \mathrm{Spec}(\mathcal{O}_{Y',y'})$ is étale, $|X'_{y'}| \to F$ is surjective, X' is affine, and $\Gamma(X', g^*\mathcal{F})$ is a free $\mathcal{O}_{Y',y'}$-module.

Proof. Choose an étale morphism $(Y', y') \to (Y, y)$ where Y' is an affine scheme. Then $X \times_Y Y'$ is quasi-compact. Choose an affine scheme X' and a surjective étale morphism $X' \to X \times_Y Y'$. Picture

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ \downarrow & & \downarrow \\ Y & \longleftarrow & Y' \end{array}$$

Then $\mathcal{F}' = g^*\mathcal{F}$ is flat over Y' at all points of $X'_{y'}$, see Morphisms of Spaces, Lemma 05VW. Hence we can apply the lemma in the case of schemes (More on Flatness, Lemma 05L1) to the morphism $X' \to Y'$, the quasi-coherent sheaf $g^*\mathcal{F}$, and the point y'. This gives an étale morphism $(Y'', y'') \to (Y', y')$ and a commutative diagram

$$\begin{array}{ccccc} X & \xleftarrow{g} & X' & \xleftarrow{g'} & X'' \\ \downarrow & & \downarrow & & \downarrow \\ Y & \longleftarrow & Y' & \longleftarrow & \mathrm{Spec}(\mathcal{O}_{Y'',y''}) \end{array}$$

To get what we want we take $(Y'', y'') \to (Y, y)$ and $g \circ g' : X'' \to X$. □